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Structural design by density compositions of poplar deadwood

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ABSTRACT: This research investigates anisotropic and highly inhomogeneous wood elements to develop new knowledge and methods for analysis, design and fabrication with complex biomaterials. Focus is placed on advanced density analysis by computer vision and computational methods, to identify structural areas for connectors between parts that allow advanced assemblies of elements, such as shingle façade systems. The studies demonstrate how materials with high density variance can be understood and used for future wood-based architectures.

1 INTRODUCTION

This research aims to understand and develop design methods for highly inhomogeneous wood characteristics with multi-modal objectives related to the structural properties of the material. The focus is placed on non-uniform element joinery, and robustness in relation to weathering impact of the building envelope, all related to the specific density properties of the material. The combination of objectives is investigated and demonstrated through a shingle façade test case by use of robotic assembly methods advanced from previous studies.

Wood is, by definition, a complex material. Each species has its particular characteristics, and each wood sample is again defined by the growth environment and the following processing. Wood is anisotropic due to cellular structure and fibre organisation and inhomogeneous due to context-influenced growth processes (Avram et al., 2023; Record, 2004; Zhang et al., 2022).

Commonly, anisotropy is attempted neutralised, or minimised, by engineering the fibres through the strategic sorting of elements (herein fibres orientation), with counteracting layers into plywood or glueing elements into glue laminated parts. The principal approach is to use wood's basic structure of fibre orientation but limit or entirely remove inhomogeneous properties and characteristics for generalisation and standardisation. Wood engineering with adhesives has provided significant advantages for laminated wood elements and timber structures, by cross-laminated timber, plywood structures and similar composites (Barbu, 2015; Foged, 2022). The flip side of this approach is that the base elements for assemblies need to be near perfect in geometry and surface properties, allowing glue-based bonding of sub-elements, leading to wasteful wood processing operations, where up to 70% of the harvested wood is downgraded to biofuel or lost during other processing of the material (Brownell II et al., 2023; Svensk Tra, 2023). With each procedural step of the chain, material is lost, and energy is used. Furthermore, the raw material used requires a high homogeneity, effectively removing the possibility in the first place of using wood material with significant defects, decay or other structural deficits related to non-uniformity.

Instead of neutralising the natural wood characteristics through processing or developing new properties through the engineering of sub-elements with adhesives, this investigation

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focuses on the increased complexity in a material caused by the decay processes of deadwood, Figure 1. The degradation of the material leads to amplified inhomogeneity (Foged, 2024), similar to materials that have been used before and degraded by prior application and weathering. By understanding how to decode complex parts of material structures, we can engage with and utilise a larger set of materials currently considered waste, both in nature and in the existing built environment. This includes wood, a scarce but important resource, which can be utilised more efficiently than current wasteful processes. The concepts and methods developed in this research address a wider potential and deficit in knowledge concerning the understanding and use of complex biogenic materials.

The performance of a material depends on its field of application, such as the ability to withstand compression force or properties of reducing heat flux to increase insulation capacities. This investigation concentrates on analysing and designing through highly local density characteristics, which is a central property of a material when joining elements together, when transferring forces between elements and how a material will deteriorate through exposure to weathering. If the material density is low, the forces resulting from connectors (such as nails or screws) between building elements may lead to rupture during assembly or earlier tearing of elements and material degradation is accelerated as rainwater, salts, wind and ultraviolet rays from sunlight will penetrate the surfaces and decrease the longevity of use.



Figure 1. Poplar deadwood with high material inhomogeneity. The material sample shown is 250x80 mm with significant transformation of the colour and texture properties through natural decay processes in the forest.

What is attempted and presented from this effort is the investigation of the material complexity and design potential through three processes of material experimentation and testing. Experimentation and testing are based on material and computational studies. The three processes include the registration of material colour and texture, then computational analysis and design with the developed material dataset and lastly the making of a test module of shingles for a façade as a demonstration and applied test case.

2 METHODS

Through a combination of analysed material samples, computational analysis and exploration, and prototyping studies, the density properties and composition of a façade structure of shingles are explored to understand and develop a combined analysis, design and fabrication methodology.

2.1 *Material studies*

Firstly, the material samples are Poplar wood from the same tree, acquired from a Danish forest in a state of significant decay following the fungal attack. This material has been chosen

due to its high inhomogeneity suitable for the study and the previous datasets developed by the authors in relation to density characteristics for integration into the developed analysis, design and fabrication methodology of this study. For the previous dataset, samples were registered with varied colours and textures, including 110 samples with the dimensions 400x100x5 mm. The sample size was chosen based on the geometrical dimensions of typical wood-based shingle systems, as seen in built works by Peter Zumthor and Bernado Bader. The surface texture and colour of the samples were captured using ordinary photographic equipment and collected as a material dataset, with a digital sample resolution of 877x2876px, each with three values of RGB/HSV colour coding, creating more than 7.5 million data points for each material sample. The new dataset of 105 samples of varying homogeneity, Figure 2 and 3, was then compared and embedded into a new computational analysis and design model (description below, section 2.2). The integration of the new dataset is then combined with the material-density dataset from previous studies (reference to be included after peer review) to evaluate and analyse the density of the new material dataset provided by this study.

Initial tests were made with photos captured with an iPhone 13 Pro, allowing for the capture and evaluation of material colours, in which RGB/HSV values are extracted and used for computational analysis and design. To ensure that the image capturing process could serve as an integrated part of the fabrication process, two additional camera technologies, more suitable for being mounted on a robot arm, were selected. The photography registrations and material analysis explored with the selected camera technologies is shown in Figure 2 and 4.

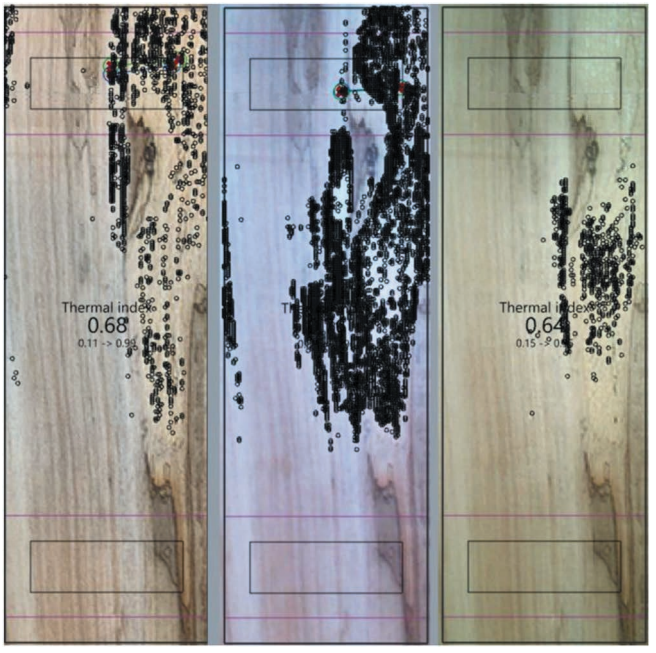


Figure 2. Material studies by use of photography registrations with iPhone 13 Pro (left), RealSense D435i (center) and RealSense L515 (right) cameras. The colour capture and representation are significantly different and influence the identification of material properties by RGB decoding.

Camera settings of *white balance*, *hue*, *brightness*, *exposure*, were empirically modified to match the observed colours of the samples to create alignment between natural colours and RGB colour representation for computational analysis and design.



Figure 3. Twenty of the 105 material samples of the poplar elements used for the shingle façade test case.

2.2 *Computational studies*

The material datasets are based on the photographic material described, with the image data captured with, and loaded directly into, a computational model based on the Rhino Grasshopper Software Framework. The image capturing process is achieved through a Python-based Grasshopper component using the Intel® RealSense™ SDK 2.0 and the Python wrapper pyrealsense2 to connect with the RealSense camera and capture data from its RGB channel. The image processing is done via the sub-program (GH add-on) Javid, enabling iterating and processing of the photographic dataset and direct writing of a new numerical dataset to MS Excel, including the analysed properties (RGB/HSV values) of each material sample. The computational analysis of each material sample is based on colour and texture identification, correlating and mapping colour properties with material densities, Figure 2. This provides a sample specific digital map of densities from which further analysis and design actions can be applied.

With the material samples intended as façade shingles, geometrical zones, marking an underlying support structure for the shingles, are positioned, Figure 2 and 4. The density analysis combined with the support structure zones is combined to filter away all areas of the samples (shingles) unsuitable for connectors to sub-structures. Within these structural support zones, point groups of high material density are identified and used for the selection of suitable connection points. With the intention to increase connectivity between the shingle and support structure, two connection points are identified. This is achieved by using two computational methods, a QuadTree algorithm and a k-nearest neighbours (KNN) algorithm, Figure 2 and 4, identifying groups of points with high density and subsequently measuring the specific distance between individual points from neighbouring groups. By ranking potential connectors based on their maximum distance the method seeks to minimize the moment of the forces acting on the connectors thereby avoiding ruptures within the shingles during construction and use.

With the structural joinery analysis and proposition process ensuring connectivity to support structures, shingles can be composed for other objectives, such as subjective visual expression, or performance-oriented goals, such as potential weathering impact by water run-off processes and exposure to UV radiation. By segmenting each shingle digitally into five groups and calculating the average density of each group in each shingle, a series of shingle density IDs are produced and stored in the numerical (MS Excel) dataset. The density markers and the IDs of each shingle are then loaded into the computational design model, enabling a designer to compose arrays of shingles that combine to produce assembly-specific properties, Figures 5 and 7.

The computational analysis and design model, operating on the material-geometrical datasets, are advanced to allow a designer to compose material-element organisations, combining intuitive qualitative-based modelling and numerical-based arrangement of building parts.

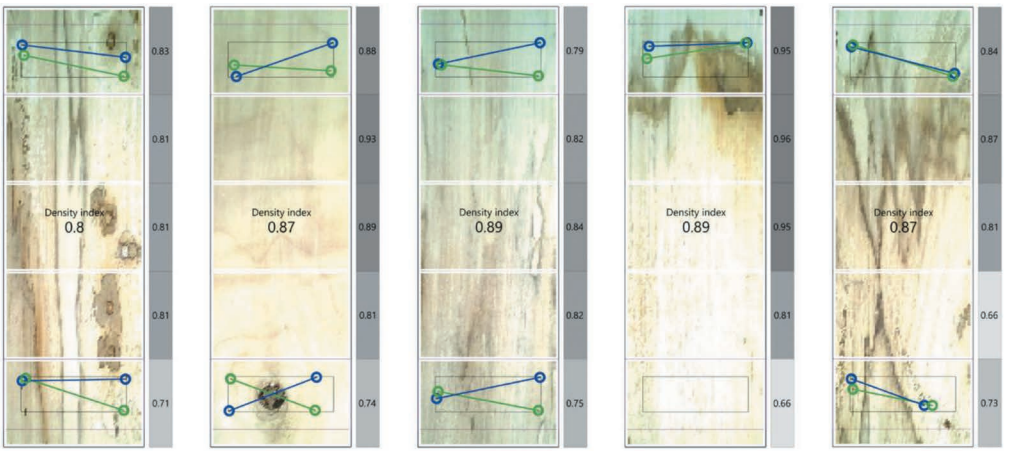


Figure 4. Computational model analysing density characteristics and proposing connection points for joinery between shingles and support structure. Blue marking is based on QuadTree and green on KNN.

In Figure 5, the two examples to the left depicts a design strategy where the composition of elements, based on their specific density IDs, is computed by increasing density from bottom > top, and left > right. The two examples to the right, show a freeform curve and polyline drawn by the designer. The intuitive drawing of lines, similar to drawing on a canvas, is connected to a proximity analysis between the drawn lines and the positions of the shingles in the building module array. From this, the density characteristics of the elements from the dataset and the positions in the building module are paired, and elements are distributed computationally as a relation between unique material properties and design intention of the building module, leading to an intuitive and explorative sorting of shingle elements, which all satisfy the connection conditions to the substructure.

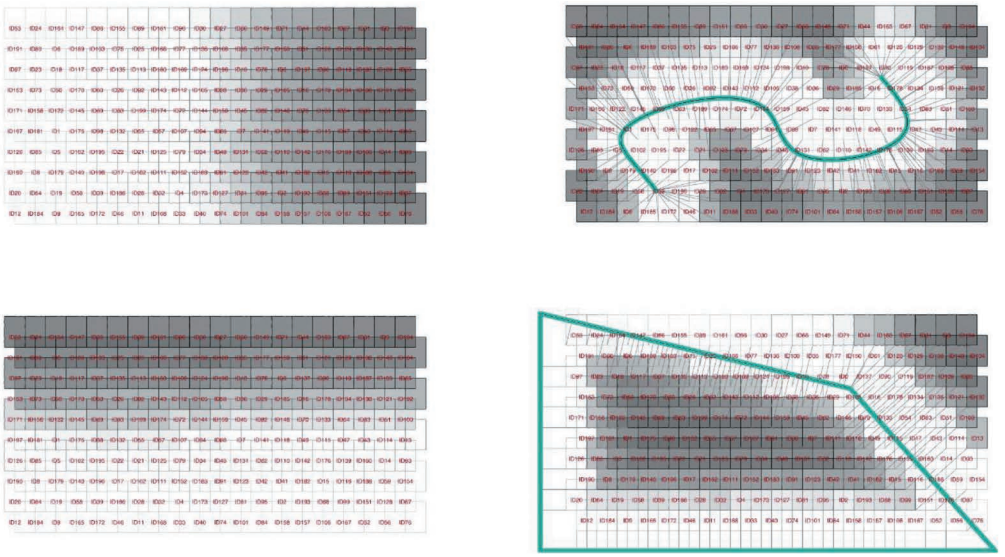


Figure 5. A computational model of 200 shingles composed by sorting the shingles based on material density data and design intent.

2.3 *Prototyping studies*

From the material and computational studies, a full building assembly of shingles is created as a prototype test of the developed methods and models. The prototype includes a wood substructure for mounting the shingles in a bespoke composition of connection points, visual expression, robustness and density characteristics as defined by the materials available and the designer's intention. The fabrication procedure is structured in three levels of robotic-aided fabrication. First level is defined by using the robot (with a mounted camera) to capture the specific shingle material properties, and after processing the density mapping described, the robot (with a mounted pen) marks on the shingle where a nail should be positioned by a human co-builder. Second level, Figure 6, is defined by using the robot arm to analyse, mark connector points and then place the shingle on the substructure, and then let the co-builder insert nails to mount the single on the connection points.

Third level is defined by using the robot arm to analyse the material, place the single on the structure and insert nails to mount the shingle on the structure, without involvement of a co-builder, making the process potentially fully automatized.

To maintain a human co-builder in the fabrication loop, the second level of robotic-aided fabrication is used during this study, Figure 6. Besides the computational analysis and design methods developed, this allows the co-builder to add layers of qualitative, ad-hoc analysis and decision making by for instance reordering the materials picked by the robot.

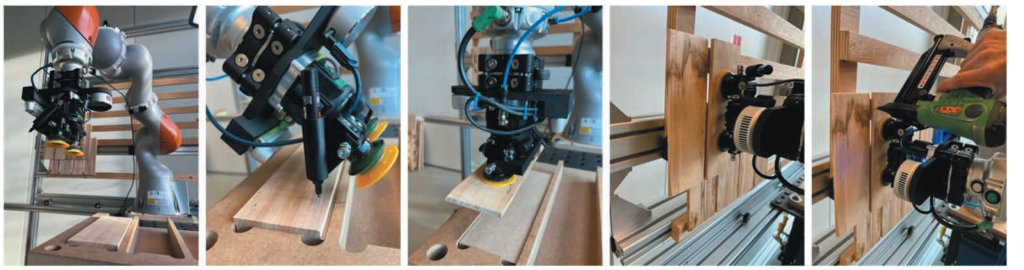


Figure 6. Fabrication method with second-level collaboration between robot system and co-builder.

3 RESULTS

The research has investigated material, computational and robotic aspects of analysing, designing and fabricating with complex wood structures as a case study for a shingle-based façade system.

Experiments based on computational studies show that the proposed method for the analysis and design processes can identify, organise and distribute shingles by their density ID and guiding principles of distribution across an array of shingle elements forming a wall segment. By combining the numerical dataset with an intuitive design interface, a combination of explorative design processes with additional decision-making information layers is achievable, making more complex positioning of shingles possible while still adhering to the binary decision processes for connector requirements to the substructure as computed by the analysis-design system.

Experiments based on material and fabrication studies show that the proposed method for the human-robot assembly of sub-elements into an array of parts is purposeful by use of robotic precision with agile human interventions of evaluating and mounting elements once the robot has analysed and positioned the elements and guided the placement of connectors, Figure 6 and 7. These results follow the testing of a level 2 human-robot collaboration, maintaining the human within the fabrication chain for quality control, possible corrective actions and as a new form for craftmanship in the intersection between automation and handmade assemblies.

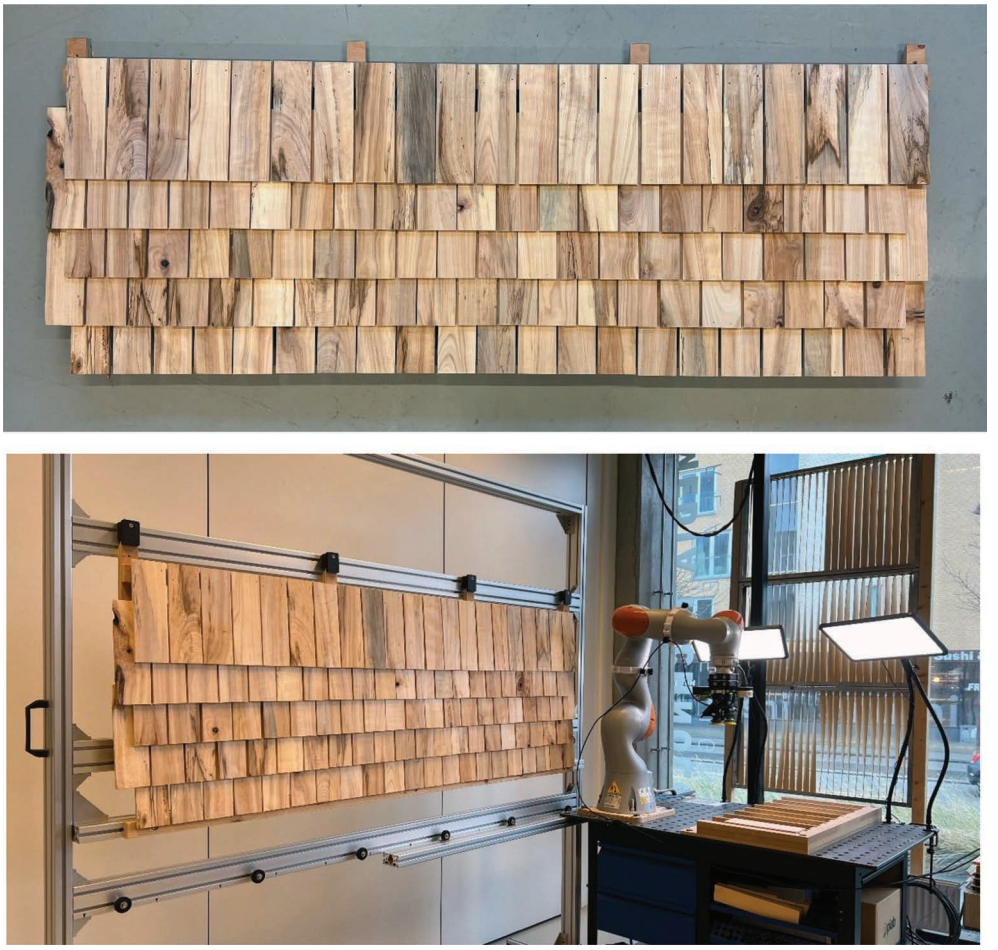


Figure 7. Wall segment with array of 105 shingles based on the developed and tested analysis-design-fabrication methodology.

4 DISCUSSION

This research uses a specified material, poplar wood in significant decay, as test samples and test cases for inhomogeneous material analysis, design and fabrication processes. The studies and findings are, therefore, tied to this specific material. However, the analysis, design and fabrication methods presented are all transferable to other materials where a material dataset is available. Applying these methods broadly requires advancements in the material datasets available, moving beyond current information levels, where materials, such as a wood species, are defined by singular, average values rather than high fidelity data structures, such as those including RBG/HSV value with correlated data for density and other material properties. Image-based machine learning models may take part in the development of larger datasets, where the registrations of physical samples can be explored and expanded by the generation of synthetic datasets. The accuracy of material properties in a synthetic data set obviously requires a high degree of validation when used for high resolution design compositions and may in some instances not be useful or directly problematic. In the context of this study, a low accuracy in the synthetic dataset would result in erroneous connector points and accelerated material degradation, which is considered impossible to detect by the designer during development.

The presented investigation of shingle-based wall segments extends upon previous research by the authors (Jensen and Foged, 2023), in which a larger façade system, consisting of 12 segments and approximately 1200 wood shingles, was constructed, showcasing the feasibility of constructing shingle-based façade systems through robotic prefabrication processes.

While traditional methods for constructing shingle facades relies on high-quality materials, this study explores a more nuanced approach that considers variation in expected climatic exposure and the resulting wear and tear of building façades. Allowing such conditional variations to inform material distribution affords a more detailed and direct relationship between expected façade exposure and material performance requirements, thereby facilitating an inclusion of materials with high density variance and inhomogeneous characteristics, resulting in the utilization of materials that are currently considered waste.

5 CONCLUSION

The study finds that the proposed method for analysis, design and fabrication of complex inhomogeneous materials can be applied to investigate, develop and design building facade compositions with raw wood material otherwise considered unsuitable for construction. Specifically, the investigation concludes that;

- The material analysis method, based on calibrated photographic analysis to match HSV/RGB values of observed colours, can identify, localise and map the density characteristics within millimetre precision of the materials used.
- The material analysis method, based on computational evaluation processes of material proximity conditions, can identify and propose positions for multiple connectors between analysed elements and an underlying structure as part of the building envelope.
- The computational methods for analysis and design provide unique propositional design processes, allowing a combination of numerical evaluation and design decision-making with qualitative interoperability by the designer for multi-modal analysis and design decision-making processes with complex inhomogeneous wood structures.
- The robotic methods for fabrication enable a linkage between advanced material analysis, material-driven design actions and fabrication procedures with multiple levels of integration.

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